

Programming for ArcGIS Desktop Add-Ins With VB.NET and ArcObjects

Chapter 1: Getting Started with Visual Basic .NET

1. Computer and programming
 - 1.1 Computer hardware
 - 1.2 Computer software
 - 1.3 How to write programs
2. Visual Basic .NET
3. Starting up VB.NET
4. Controls, properties, and events
 - 4.1 Control properties
 - 4.2 Control events
 - 4.3 Setting property values by code
5. Simple Windows Applications
6. Understanding files of a VB.NET project

Chapter 2: VB.NET Variables and Data Types

1. Variables
2. Data types
3. Arithmetic operations
 - 3.1 Operators
 - 3.2 Incrementation
 - 3.3 Associativity
4. Built-in math functions
5. Strings
 - 5.1 String variables and string values
 - 5.2 String concatenation
 - 5.3 Using text boxes for input and output
 - 5.4 Conversion between strings and numbers
 - 5.5 String properties and methods
6. Three types of errors
7. Additional techniques
 - 7.1 Message box
 - 7.2 Comments
 - 7.3 Code line breaking (Line continuation)

Chapter 3: Flow Control and Debugging

1. ASCII
2. Relational and logical operations
3. Decision
 - 3.1 If block
 - 3.2 Select Case block
4. Repetition
 - 4.1 For ... Next repetition

- 4.2 Nested repetitions
- 4.3 Do ... Loop and While ... End While
- 5. Debugging
 - 5.1 Breakpoint
 - 5.2 Stepping through a program
 - 5.3 Examining variable values

Chapter 4: Procedures

- 1. Functions
- 2. Sub procedures
- 3. Program flow with procedure calls
- 4. Scope of variables
- 5. Passing arguments ByVal vs. ByRef
 - 5.1 ByVal
 - 5.2 ByRef

Chapter 5: Arrays and Text Files

- 1. Array
- 2. Reading text files
- 3. VB.NET error handling
- 4. Open file dialog
- 5. Resizing arrays: ReDim and Preserve
- 6. Parsing strings
- 7. Stream reader and stream writer
 - 7.1 The StreamReader class
 - 7.2 The StreamWriter class

Chapter 6: Object-Oriented Programming

- 1. Structure
- 2. Object-Oriented programming
 - 2.1 Encapsulation
 - 2.2 Inheritance
 - 2.3 Polymorphism
- 3. Class structure
- 4. Example: PointCity Project
 - 4.1 Creating classes and objects
 - 4.2 Property structure
 - 4.3 Inheritance
 - 4.4 Polymorphism
- 5. Namespace
- 6. Enumeration

Chapter 7: Building ArcGIS Desktop Add-Ins

- 1. ArcGIS Desktop Add-In
- 2. Creating a new ArcMap add-in project
- 3. Using the ArcGIS add-Ins wizard to create buttons and tools

4. Understanding the XML
5. Implementing functionality of the add-in button
6. Implementing functionality of the add-in tool
7. Creating a custom combo box
8. Creating a custom toolbar

Chapter 8: Understanding Object Model Diagrams

1. About ArcObjects
2. UML for class diagrams
 - 2.1 Class structure
 - 2.2 Class relationships: association, inheritance, aggregation, and instantiation
3. Concepts and techniques of ArcObjects programming
 - 3.1 Component Object Model (COM)
 - 3.2 Multiple interfaces
 - 3.3 Query Interface (QI)
 - 3.4 Class types: co-class, abstract class, and regular class
 - 3.5 Class inheritance and interface inheritance
4. ArcObjects assemblies and namespaces

Chapter 9: Accessing Map Layers

1. Application and MxDocument
2. The Carto Assembly
3. Turning off of a layer
4. Importing Namespaces
5. Turning on/off all layers
6. Changing map scales
7. Obtaining coordinates of mouse location

Chapter 10: Loading Layers and Rendering Feature Layers

1. Loading Feature Layers
 - 1.1 Model diagram analysis and solution development
 - 1.2 Creating an add-in button
 - 1.3 Implementing the load layer button
 - 1.4 Enhancement
2. Loading raster layers
 - 2.1 Object model diagram analysis and solution development
 - 2.2 Creating a function to load raster layers
3. Rendering feature layers
 - 3.1 Model diagram analysis and solution development
 - 3.2 Creating a simple fill renderer

Chapter 11: Manipulating Elements and Geometry

1. Creating marker elements
 - 1.1 Model diagram analysis and solution development
 - 1.2 Implementing the bus stop tool
2. Creating line elements

- 2.1 Object model diagram analysis and solution development
- 2.2 Implementing the bus route creation tool
- 3. Displaying GPS data as line elements
 - 3.1 Creating an add-in dockable window
 - 3.2 Creating polyline objects from coordinates
 - 3.3 Retrieving coordinates from a GPS data file
 - 3.4 Final assembly
 - 3.5 Clearing elements from the map

Chapter 12: Selecting Features

- 1. Working with multiple data frames (maps)
 - 1.1 Setting a name to the focus map
 - 1.2 Creating new map objects in a map document
 - 1.3 Accessing and activating a map
- 2. Definition query
- 3. Selecting features
- 4. Selection sets and cursors
 - 4.1 SelectionSet
 - 4.2 Cursor
- 5. Feature cursors and features

Chapter 13: Manipulating Feature Data

- 1. Accessing attribute values
 - 1.1 SpatialFilter
 - 1.2 Fields and Value
 - 1.3 Creating your own identify tool
- 2. Editing attribute values
 - 2.1 Problem analysis
 - 2.2 Implementation
- 3. Adding new features into feature classes
 - 3.1 Manipulating a geodatabase
 - 3.2 Adding new features into a feature class
 - 3.3 Final assembly

Chapter 14 Manipulating Raster Data

- 1. Identifying pixel values
- 2. Creating new raster datasets
 - 2.1 Raster dataset
 - 2.2 Raster and pixel block
 - 2.3 Implementation
- 3. Local function - raster reclassification
 - 3.1 Object model analysis and solution development
 - 3.2 Implementing
- 4. Focal function - slope raster generation
 - 4.1 Slope algorithms
 - 4.2 Handling no data value
 - 4.3 Getting processing windows from a pixel array

- 4.4 Overall procedure
- 4.5 Final assembly

Chapter 15 Using Geoprocessing Tools

- 1. Geoprocessing and geoprocessing tools
- 2. Toolbox assemblies
- 3. Tool classes
- 4. The Geoprocessor
 - 4.1 Setting geoprocessing environment
 - 4.2 Executing geoprocessing tools
 - 4.3 Monitoring geoprocessing execution status
- 5. Application
 - 5.1 Analysis Tools
 - 5.1.1 Select tool
 - 5.1.2 Clip tool
 - 5.1.3 Identity tool
 - 5.1.4 Buffer tool
 - 5.2 Spatial Analyst Tools
 - 5.2.1 Slope tool
 - 5.2.2 Contour tool
 - 5.3 Data Management Tools

Chapter 16 Application Development: Data Driven Pages

- 1. Introduction to data driven pages
- 2. Creating index layer
 - 2.1 Defining mapping extent
 - 2.2 Creating a blank index feature class
 - 2.3 Creating tile features
 - 2.4 Editing tile features
 - 2.5 Generating necessary fields: page names, page number
 - 2.6 Generating tile neighbors
 - 2.7 Setting up data frames and layers
- 3. Page layout design
 - 3.1 Adding map elements
 - 3.2 Adding map surrounds
- 4. Enabling data driven pages